



Water Monitoring and Truck Tracking Web System

Vibe Tech business proposal:



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Introduction

The Northern Cape experiences recurring water Shortages due to various reasons such as drought and lowered dam levels as seen in the local news video showing residents of Ritchie in the Northern Cape struggling to secure reliable water from the Salt Blight municipality, highlighting the human impact of failing water systems.

This problem has given a rise the water crisis in the Northern Cape such that Communities in the northern Cape are becoming vulnerable as they receive little to no early warning before disruptions occur. This project aims to create a web-based system to help residents not be vulnerable during these times.

Problem Statement

Municipalities in the Northern Cape face significant challenges in managing and communicating water supply services. Communities often experience sudden water supply interruptions without receiving early warnings to alert them about impending shortages. Public access to dam level information is limited making it difficult for residents and local administrators to anticipate water availability issues.

Communication about water restrictions is often not clear or delayed which further worsens the situation. In addition, there is no real-time system that tracks water truck deliveries, and this leads to missed or delayed deliveries and a lack of accountability in distribution services. These challenges negatively affect residents, municipal water departments, water truck drivers and local administrators. Currently, there is no centralized web-based system that is particularly designed for municipalities that integrates water monitoring, shortage prediction and delivery tracking to enhance transparency, efficiency and service reliability.

Existing solution

At present when there is a shortage of water or when the piped water systems stop working the municipality rely on water tanker trucks to supply water to the communities in the Northern Cape. According to national statistics about 27.3% of

South African households depend on water tankers when water from the pipes pipe is unavailable for long periods.

Municipalities normally relay information about water shortages and when the water tank will deliver water through public social media posts, local radio stations, or community leaders such as councillor.

Water tanker deliveries are generally managed through manual coordination by municipal water departments. This often involves phone calls, paper records, or simple scheduling methods to send trucks to different areas.

These existing methods however have a lot of limitations. Communication about water interruptions is often delayed or unclear, which leaves residents unprepared for water shortages. In addition, there is no real-time tracking system for water trucks, making it difficult for residents to know when or where water tank trucks will arrive. This system relies heavily on manual processes and deliveries may sometimes be delayed because it is inefficiently scheduled.

As a result of this the current solution lacks transparency, co-ordination and reliable information sharing. This highlights the need for a centralized web-based system that can improve communication, monitor water availability, and track tanker deliveries more effectively.

PROPOSED SOLUTION

Our proposed solution is to develop a web-based application that integrates real-time water monitoring with water truck tracking to support municipalities in the Northern Cape. The system will provide residents with secure login and user registration, allowing them to view real-time dam water levels, historical water level trends, water conservation tips, scheduled water restrictions, real-time truck locations (GPS simulation), area-based delivery schedules, and receive early alert warnings. At the same time, the application will allow municipalities to manage and monitor water truck deliveries through live tracking, delivery scheduling, and performance reporting, improving transparency and accountability.

System Architecture



Presentation Layer (User Interface-Front-end)

- **Resident Portal:** Allows rural residents to log in securely, view real-time dam levels and historical trends, see scheduled water restrictions, track water truck locations via a simulated GPS map, view delivery schedules, and submit or track complaints.
- **Admin (Municipality) Portal:** Used by local administrators and water departments to update dam data, upload historical data, generate risk analyses, manage water trucks, assign routes, monitor deliveries, and view an analytics dashboard of high-risk zones.

Application Layer (Back-end)

- **Authentication and User Management:** This will secure login and user registration for residents, admins, and drivers.
- **Water Monitoring and Analytics:** This processes real-time and historical dam water level data. It is responsible for generating water-shortage risk analyses and creating monthly distribution reports.
- **Notification and Alert:** Automatically triggers early warning shortage alerts and sends SMS and email notifications to residents based on predictive data.
- **Complaint Management System:** Receives resident complaints regarding missed deliveries and tracks their resolution status.

Data Layer (Database)

- **User Database:** Stores credentials and profiles for residents, administrators, and drivers.
- **Water Data Repository:** This Stores real-time dam levels, historical water data, and water conservation tips.
- **Logistics Database:** Stores truck details, delivery routes, schedules, and driver performance records.
- **Feedback Database:** Stores residents' complaints and their current tracking status.

External Integrations (APIs)

- **GPS/Mapping Simulation API:** Feeds real-time location data to the web app so residents and admins can visually track the water trucks.
- **SMSC - SMPP Gateway:** Communicates with the Notification Engine to send out the actual SMS notifications for water warning.
- **Email Gateway/SMTP:** Communicates with the Notification Engine to send out the actual email notifications for predicted shortages.